

Exercise 7 – Time series LSTM GRU

Objective: Implement GRU and Recurrent Neural Networks (RNNs) in Python. Students will choose one of the two datasets and will perform **time series prediction** .

1. Dataset Selection:

- a. Select a dataset from the list below (or find another relevant dataset with instructor approval):

Time Series Datasets:

- i. [Electricity Load Forecasting Dataset](#)
- ii. [Kaggle Stock Market dataset](#)

2. Build the Model:

- a. Implement an LSTM GRU neural network in Python (using PyTorch based on the notebook practice).
- b. Include the following:
 - i. Input preprocessing (e.g., tokenization for sentiment analysis or scaling for time series).
 - ii. LSTM architecture.
 - iii. Training and validation process.

3. Train and Test the Model:

- a. Train your LSTM model using the dataset.
- b. Evaluate its performance using relevant metrics (e.g., RMSE for time series, accuracy/F1-score for sentiment analysis).

4. Write a Report:

- a. Submit a report detailing:
 - i. Dataset description and preprocessing steps.
 - ii. LSTM architecture and training details.
 - iii. Results and analysis (include plots of training/validation loss and performance metrics).
 - iv. Conclusions and insights (e.g., challenges, potential improvements, future applications).

Deliverables:

1. Python code implementation in a Jupyter Notebook or script.
2. A detailed report (PDF or Word document) covering all required points.

